Chapter 22

Volatility Products: VIX, Variance Swaps, and Dispersion

Section 12.4 said that a trader who wants volatility exposure must delta-hedge an option and that the residual is a bet on realized against implied. Chapter 18 came within a step of saying what that residual is: a delta-hedged position is long gamma and short theta, and Section 18.4 found that it comes out level on a day the stock moves by one daily standard deviation at its implied volatility. Section 22.1 takes the step. Both descriptions make volatility something you get *left with* after removing what you did not want.

This chapter is about the instruments that deliver it directly. A variance swap pays realized variance against a fixed rate with no delta to hedge and no dependence on where the price moved. A VIX futures contract references an index computed from option prices. A dispersion trade isolates correlation. Each of these turns a by-product into a contract, and each is imperfect in a different and instructive way.

Why anyone wants such contracts shapes the whole market. Everyone who holds equities is short volatility whether they know it or not: drawdowns arrive when volatility rises, and the hedges that would soften them cost the most at exactly that moment. So there is a natural buyer of volatility, the holder of equity, and a natural seller, the intermediary or fund paid to bear what the holder wants to shed. Section 19.10 put the price of that transfer at two to four points of implied over realized volatility on equity indices and called it the volatility risk premium. The instruments here let that premium be bought and sold on its own, without the delta, the strike and the rebalancing that come with an option. That is their appeal. It is also why the sellers tend to be crowded, and why, with investors in aggregate structurally short volatility through their equity, the episodes in this chapter (2008, February 2018, March 2020) recur in the same shape. The route follows the order in which the market built the instruments: the variance swap and its caps (Sections 22.2 to 22.5), VIX, its futures and its products (Sections 22.6 to 22.8), then the dealer's view (Sections 22.9 to 22.13).

22.1 THREE WAYS TO OWN VOLATILITY

Delta-hedged options. Buy an option and hedge the delta. Section 18.1's dealer, hedging a written call, stood on the other side of this trade. The hedge cancels the delta term of Chapter 18's expansion, equation (18.1), and what is left are its gamma and theta terms: the buyer earns $\frac{1}{2}\Gamma(\Delta S)^2$ on each day's move and pays theta, whose cost over the day, with the interest term dropped, is $\frac{1}{2}\Gamma S^2\sigma^2\,\Delta t$ at the implied volatility σ. With the day's return written $\Delta S/S$, the day's profit is

$$\frac{1}{2}\Gamma S^2\Big[(\Delta S/S)^2-\sigma^2\,\Delta t\Big]$$

realized less implied variance, weighted by $\frac{1}{2}\Gamma S^2$. Summed over the position's life this is Section 12.4's residual, but weighted by gamma: it depends on *where* the underlying moved, not only how much, and two paths with the same realized volatility pay differently. The identity

holds for every option (a put, a butterfly, the straddle traders use because it starts close to delta-neutral); only Γ changes. It also holds σ fixed: if implied volatility moves, the vega term of equation (18.1), Section 18.5's vega profit or loss, is added.

Variance swaps. Contract directly on realized variance. There is no delta, no gamma weighting and no rebalancing. The exposure is exactly what the name says.

VIX futures. Trade an index constructed from option prices. This is the most accessible and the least direct: a VIX futures contract is not a claim on volatility but a claim on the *level of a computed index at a future date*, and Section 22.7 shows how far apart those can be.

The three are not substitutes, and most of the trouble in this market comes from treating them as though they were.

22.2 THE VARIANCE SWAP

A variance swap pays, at maturity,

$$N_{\text{var}}\left(\sigma_R^2 - K_{\text{var}}^2\right) \tag{22.1}$$

where σ_R^2 is the annualized realized variance of log returns over the life of the contract, K_{var} is the strike, quoted as a volatility, and N_{var} is the variance notional, paid per variance point (volatilities in percentage points, so a realized variance of 0.04 enters as $20^2 = 400$).

The vega notional convention. Variance notionals are awkward to think in, so the market quotes **vega notional**, the slope of the payoff in realized volatility, taken at the strike, so that a one-point move in realized volatility earns N_{vega} to first order. Since the slope of (22.1) at $\sigma_R = K_{\text{var}}$ is $2K_{\text{var}}N_{\text{var}}$,

$$N_{\text{var}} = \frac{N_{\text{vega}}}{2K_{\text{var}}} \tag{22.2}$$

so that, by (22.2), the payoff (22.1) becomes $N_{\text{vega}}(\sigma_R^2 - K_{\text{var}}^2)/(2K_{\text{var}})$. A desk saying it is "short \$100,000 of vega at 20" means it loses roughly \$100,000 if realized volatility comes in at 21 rather than 20—*roughly*, because the payoff is not linear in volatility: the exact loss is $100{,}000 \times (21^2 - 20^2)/(2 \times 20) = \$102{,}500$, and the extra 2.5% is the convexity that is the subject of Section 22.4.

Realized variance is defined, not observed. The contract specifies the observation times (daily closes), the return convention (log returns), the annualization factor (252), and, importantly, whether the mean is subtracted:

$$\sigma_R^2 = \frac{252}{n}\sum_{i=1}^{n}\left(\ln\frac{S_i}{S_{i-1}}\right)^2 \tag{22.3}$$

with S_0 the close on the trade date and S_i the official close on the ith of the n scheduled observation dates. That sum is the contract. It is what the confirmation says and what settlement is computed from, and no integral appears in any document. No mean is subtracted in it, and that is standard practice: it is what lets the option strip of Section 22.3 replicate the payoff, and it makes the figure differ slightly from a sample variance. These are not details: two contracts with the same strike and different terms are different contracts, and the cap that Section 22.5 describes, standard on single-stock contracts and found on index contracts since 2008, is one such term.

22.3 REPLICATING VARIANCE

The result that makes this market work is that variance can be replicated from vanilla options held *statically*: no model, no volatility forecast, and nothing traded as the price moves except a stock position of constant dollar size. The route is Chapter 12's. Section 12.8 built any

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piecewise-linear payoff from a bond, a forward, and options at its kinks; a smooth payoff is the limit of piecewise-linear ones, so it is built the same way from a continuous strip of options. Apply that to the smooth payoff $-2\ln(S_T/F)$, the **log contract**, with T the contract's maturity in years and F today's forward price for that date. Two facts combine, and each takes a paragraph.

Delta-hedging a log contract earns exactly realized variance. Let the price follow

$$\frac{dS}{S} = \mu_t\,dt + \sigma_t\,dW$$

with μ_t any drift and σ_t any volatility process (constant, stochastic, path-dependent); nothing below depends on which. Itô's lemma applied to $\ln S$ gives

$$d(\ln S) = \Big(\mu_t - \frac{1}{2}\sigma_t^2\Big)dt + \sigma_t\,dW, \qquad \text{so} \qquad \frac{dS}{S} - d(\ln S) = \frac{1}{2}\sigma_t^2\,dt$$

The gap between the simple return and the log return accumulates half the variance, whatever the drift and whatever the path. Integrating over the life of the contract and multiplying by $2/T$,

$$\frac{1}{T}\int_0^T \sigma_t^2\,dt = \frac{2}{T}\left[\int_0^T \frac{dS}{S} - \ln\frac{S_T}{S_0}\right] \tag{22.4}$$

The left side is realized variance, in continuous time: the quantity the contract's sum (22.3) approximates with daily closes, and the limit that sum tends to as the observations become dense, by the $dW^2 = dt$ of Section 14.7. The right side is a trading strategy: the integral is the profit, before financing and dividends, of holding $1/S_t$ of the stock at every instant (a continuously rebalanced position with a constant dollar exposure), and the logarithm is the payoff of a contract on $\ln S_T$. Realized variance is therefore two things a desk can hold, and no probability has entered.

The gamma of the log contract does the same job one rebalance at a time. Its payoff is $-2\ln(S_T/F)$, and its value at time t, V_t, is that payoff's risk-neutral expectation discounted to t. Take σ constant for the moment and write $F_t = S_t e^{(r-q)(T-t)}$ for the forward at t, with r the risk-free rate and q the dividend yield. Under the risk-neutral measure $\mu_t = r - q$, and integrating the $d(\ln S)$ equation above from t to T gives $\ln S_T$ the mean $\ln F_t - \sigma^2(T-t)/2$, so

$$V_t = e^{-r(T-t)}\,E_t^Q\left[-2\ln\frac{S_T}{F}\right] = e^{-r(T-t)}\left[-2\ln\frac{F_t}{F} + \sigma^2(T-t)\right]$$

and its gamma is

$$\Gamma = \frac{\partial^2 V_t}{\partial S^2} = \frac{2e^{-r(T-t)}}{S^2}$$

which is the payoff's own second derivative, $2/S^2$, multiplied by the discount factor. The dividend yield sits inside F_t and does not reach the second derivative. The daily gamma gain of Section 18.3 is therefore

$$\frac{1}{2}\Gamma(\Delta S)^2 = \frac{1}{2}\cdot\frac{2e^{-r(T-t)}}{S^2}\,(\Delta S)^2 = e^{-r(T-t)}\Big(\frac{\Delta S}{S}\Big)^2$$

the present value of the day's squared return, whatever the level of S. That is precisely the weighting that makes the summed gamma gains independent of where the moves happened: an ordinary option's gamma is concentrated near its strike, and the same move earns it more there than elsewhere. Carried to expiration at the risk-free rate, each day's gain is worth exactly the squared return, and that is the e^{rT} in (22.5) below: the strip's option prices are present values, and the replication multiplies them back up.

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The log contract's payoff is replicated by a strip of options weighted $1/K^2$. So the contract can be built from what trades. Section 12.8 built piecewise-linear payoffs by placing an option at each kink, in a quantity equal to the change in slope there. A smooth payoff $f(S_T)$ is the limit: a kink at every strike, with slope change $f''(K)\,dK$. Expanding around the forward, with puts below it and calls above so that every option used is out of the money,

$$f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)\,(K - S_T)^+\,dK + \int_F^\infty f''(K)\,(S_T - K)^+\,dK$$

For $f(S_T) = -2\ln(S_T/F)$ the pieces are

$$f(F) = 0, \qquad f'(F) = -\frac{2}{F}, \qquad f''(K) = \frac{2}{K^2}$$

a short position in $2/F$ forward contracts, which costs nothing to enter, and $2\,dK/K^2$ of an option at every strike.

Write σ^2_{var} for the fair variance swap rate: the strike K^2_{var} at which (22.1) is worth zero today, which is the risk-neutral expectation of realized variance, $\sigma^2_{\text{var}} = E^Q[\sigma^2_R]$. Take that expectation of (22.4). Under the risk-neutral measure the stock's expected return is $r - q$, so the stock position's expected profit is

$$E^Q\left[\int_0^T \frac{dS}{S}\right] = (r - q)\,T = \ln\frac{F}{S_0}$$

and combining it with the log term gives

$$\sigma^2_{\text{var}}\,T = -2\,E^Q\left[\ln\frac{S_T}{F}\right]$$

so the fair total variance $\sigma^2_{\text{var}}T$ is the expected payoff of the log contract. Take that expectation term by term in the expansion above: the forward term has zero expectation, and each option's expected payoff is its price grown at the risk-free rate, $e^{rT}P(K)$ or $e^{rT}C(K)$, with $P(K)$ and $C(K)$ today's prices of the put and call struck at K and expiring at T. The result is

$$\sigma^2_{\text{var}} = \frac{2e^{rT}}{T}\left[\int_0^F \frac{P(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\right] \tag{22.5}$$

which is (19.2), now derived rather than asserted. Figure 22.1 shows what the $1/K^2$ weighting looks like across the strip: the low strikes carry most of the weight (a strike at 50 four times the at-the-money strike's, a strike at 200 a quarter of it), which is why a variance swap is far more sensitive to the downside wing than a trader used to at-the-money vega would expect.

Example 22.1 Building the strip, and what truncation costs

Take a flat 20% surface, one year, $S_0 = 100$, $r = 2\%$, no dividends (forward 102.02); the answer must be exactly 20%.

Build the strip from options out of the money relative to the forward, at strikes every 5 points from 50 to 200, weight each by $\Delta K/K^2$, and apply (22.5):

$$\text{replicated volatility} = \mathbf{19.96\%}$$

against a true 20.00%. The obvious suspect for the 4 basis points of error is the truncation of the integral at 50 and 200, and the table shows it is the wrong one.

The table widens the range and, in its last row, narrows the spacing:

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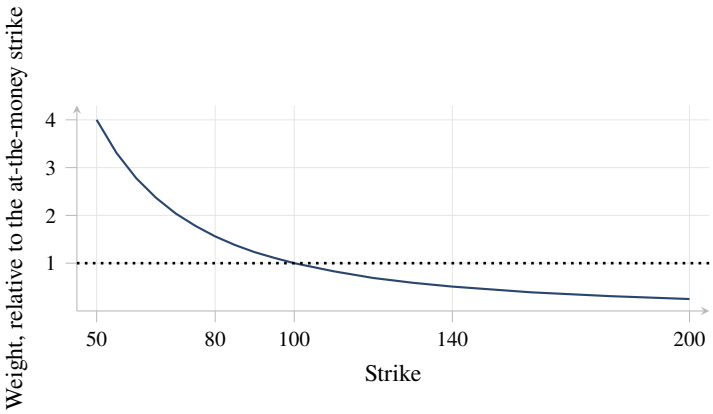


Figure 22.1: Weight on each option in the variance strip, relative to the at-the-money strike ($S_0 = 100$).

Strike range and spacing	Replicated volatility
50–200, step 5	19.956%
30–300, step 5	19.956%
20–500, step 5	19.956%
10–800, step 2	20.015%

Two lessons follow. Widening the range beyond 50–200 buys nothing, because at a 20% volatility a one-year put struck at 50 is worth a fraction of a cent and contributes nothing despite its large weight. But the spacing matters, and not because the integrand is hard. The error depends on where the grid meets the forward, 102.02, at which the strip switches from puts to calls and the out-of-the-money price peaks in a kink: a strike sitting on that peak credits its whole cell with the peak value, so a strike exactly on the forward overshoots by +10 basis points, while this 5-point grid, straddling it, falls 4 short. The error shrinks with the square of the spacing, so the 2-point grid, whose strike at 102 is close to that grid's worst placement, is within 1.5 basis points.

The ranking of the two errors reverses when volatility is high. At 60% volatility the options beyond 50 and 200 are worth something real, the truncation error dominates, and Section 19.3's warning applies with force: the strip stops where the quoting rule stops, not where the integral does.

22.4 WHY VARIANCE AND NOT VOLATILITY

The obvious contract would pay realized *volatility*, which is what people think in. The market trades variance instead, and the reason is Section 22.3: variance is replicable and volatility is not. Volatility swaps exist, and trade mostly in foreign exchange, where they are often quoted in place of variance swaps, but they are priced off variance swaps with an adjustment, and Example 22.2 shows that the adjustment is large enough to matter.

Example 22.2 The variance–volatility spread

Suppose realized volatility over the period is, under the risk-neutral measure, lognormal with mean 20%. The variance swap strike is $\sqrt{E^Q[\sigma_R^2]}$ and the volatility swap strike is $E^Q[\sigma_R]$, and by Jensen's inequality the first exceeds the second.

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Vol-of-vol ν	Variance swap	Volatility swap	Spread
20%	20.40%	20.00%	0.40 pts
30%	20.92%	20.00%	0.92 pts
50%	22.66%	20.00%	2.66 pts
70%	25.55%	20.00%	5.55 pts

Since for a lognormal $E^Q[\sigma_R^2] = (E^Q[\sigma_R])^2 e^{\nu^2}$, the spread is $\bar{\sigma}\left(e^{\nu^2/2} - 1\right)$, where $\bar{\sigma}$ is the mean of realized volatility over the period and ν the standard deviation of its logarithm over the same period (a vol-of-vol measured over one period), so the spread grows quickly with ν. If the period's realized volatility has a log standard deviation of 40% to 60%, as is realistic for a one-year index swap, the variance swap strike sits one and a half to four volatility points above the volatility swap strike on the same underlying.

Two consequences follow. A variance swap strike is not a volatility forecast. Quoting a 22.66% variance swap and calling it "the market's expected volatility" overstates the forecast by 2.66 points. This is not a further overstatement on top of the excess over at-the-money volatility that Section 19.3 attributed to the skew: at-the-money implied volatility sits close to the volatility swap strike (a little below it when spot and volatility move inversely), so that excess is largely this same gap.

And the spread is itself a traded view. Buying a volatility swap against a variance swap is a position in vol-of-vol, one of the few ways to trade it without options on variance.

22.5 THE CONVEXITY IS NOT FREE

The same convexity that makes the strike high makes the payoff asymmetric, and Figure 22.2 shows it directly: the variance swap's payoff curves away from the straight line a volatility swap would pay, touching it at the strike and lying above it everywhere else. The gap grows with the square of the distance from the strike, equally on both sides, but realized volatility can rise much further above the strike than it can fall below it.

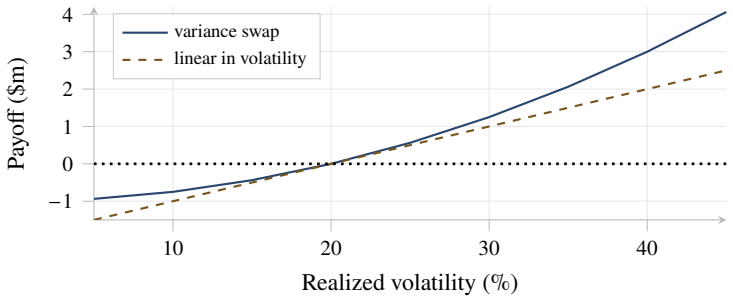


Figure 22.2: Payoff of a long variance swap struck at 20%, vega notional $100,000 per point.

Example 22.3 The buyer's convexity, and the seller's problem

A long variance swap struck at 20% with $100,000 of vega notional pays:

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Realized	Payoff	If linear in vol	Convexity gain
10%	−$750,000	−$1,000,000	+$250,000
15%	−$437,500	−$500,000	+$62,500
20%	$0	$0	$0
25%	+$562,500	+$500,000	+$62,500
30%	+$1,250,000	+$1,000,000	+$250,000
40%	+$3,000,000	+$2,000,000	+$1,000,000

The buyer gains from convexity in both directions, which is why the strike is above the volatility swap's. The seller pays for it in one direction only in practice, because realized volatility cannot fall below zero but can rise without limit: at 40% realized the seller loses $3 million on a position whose best realistic gain is the $750,000 at 10%. The seller's loss has no upper bound, while its gain can never exceed $1 million, the payoff at zero realized volatility.

In 2008 realized volatility on equity indices reached levels several times the strikes at which variance had been sold, and sellers, many of them dealers who treated the short variance as an offset to the options they were long, took losses far beyond anything the vega notional suggested.

The market's response was structural. A **cap** on realized volatility, conventionally 2.5 times the strike, had been standard on single-stock variance swaps from the start, because their gaps make the tail far worse; after 2008 that market nearly stopped trading even with its caps, and capped index contracts appeared too, though uncapped ones still trade. A capped variance swap is a variance swap less a call on realized variance struck at the cap, so it is no longer statically replicable by (22.5); the cap must be priced with a model of how forward variance moves, such as the rough-volatility models of Section 21.8.

So the instrument that was attractive because it needed no model acquired one, as a direct result of the payoff its replication made possible.

Case Study 22.1 March 2020, and what a cap is worth

The 2008 lesson of Example 22.3 was tested twelve years later, and where the repair had been written into the contract, it held. Realized volatility on major equity indices ran above 80% annualized over the worst month of the pandemic sell-off, against variance swap strikes of 12% to 18% set in the weeks before it. On uncapped terms (the pre-2008 terms, on which much of the index market still traded), that is a loss no seller's capital was sized for.

Take a one-month swap struck at 15% just before the sell-off began, with $100,000 of vega notional, and suppose realized volatility over its life came in at 80%.

Uncapped, the seller loses

$$100{,}000 \times \frac{80^2 - 15^2}{2 \times 15} = 100{,}000 \times 205.83 = \$20.6 \text{ million}$$

on a position whose stated vega notional was one hundred thousand dollars. That factor of 206 is the whole problem with the instrument, and it involves no model error—it is what the payoff formula does when realized volatility comes in at more than five times the strike.

Capped at 2.5 times the strike, the loss stops at

$$100{,}000 \times \frac{37.5^2 - 15^2}{2 \times 15} = 100{,}000 \times 39.375 = \$3.9 \text{ million}$$

The cap reduced the loss by a factor of **5.2**, and, more to the point, it converted an unbounded exposure into a known one that could be capitalized against in advance. Two observations follow.

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The repair was structural, not behavioral. No one forecast the pandemic. What the market did after 2008 was change the contract so that a failure to forecast would not be fatal, which is a different and more reliable kind of prudence.

The cost was paid in model risk. Example 22.3 noted that a capped variance swap is no longer statically replicable. The sellers of capped variance who came through March 2020 with bounded losses did so holding an instrument they had to price with Chapter 21's machinery rather than with a strip of options. They traded a known unbounded risk for a smaller risk they could only estimate, and on this occasion that was clearly the right trade.

22.6 VIX: AN INDEX, NOT A PRICE

VIX is the variance swap rate for a 30-day maturity on the S&P 500, expressed as a volatility. But the published number is not (22.5). It is a discrete sum over a strip that Cboe, the exchange that publishes the index, selects by a rule fixed in advance, and every peculiarity of the index is a property of that rule rather than of the theory: why it sits above at-the-money implied volatility, why it depends on how thickly the wings are quoted, why it can move on a day when no volatility changed. So the rule is worth writing out.

The forward comes from the options, not from the index. For each expiration, with T its time to expiration and r the matching risk-free rate, let K^* be the strike at which the call and the put are closest in price. The construction sets

$$F = K^* + e^{rT}\big(C(K^*) - P(K^*)\big) \tag{22.6}$$

which is put–call parity read backward. Parity holds at every strike, so any listed strike would return the same F from error-free quotes; Cboe uses K^* because there both options are near the money and liquid, and the quote error in $C - P$ is smallest. K^* is a choice of estimator, not a requirement of the identity. Neither the S&P 500's level nor a dividend forecast enters. Whatever the market believes about carry over the next month is already inside the option prices, and (22.6) extracts it rather than assuming it, so the forward around which VIX is built is not the S&P 500's quoted level.

The strip is the out-of-the-money options around K_0. Let K_0 be the highest listed strike at or below F. The strip takes puts at strikes below K_0, calls above, and at K_0 itself the average of the two, so that every option used is out of the money except the pair at K_0, whose call is slightly in the money because $K_0 \le F$, and none is used twice. Each carries ΔK_i, half the distance between its neighbors. Walking outward from K_0 in each direction, the calculation stops after two consecutive strikes with no bid. Writing $Q(K_i)$ for the bid–ask midpoint of the option used at K_i (Cboe's notation, not the measure), the expiration's variance swap rate σ^2 (σ^2_{var} of Section 22.3) is

$$\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{rT} Q(K_i) \;-\; \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 \tag{22.7}$$

The sum is (22.5) discretized. The subtracted term is not, and it is the one piece of the method that does not follow from Section 22.3.

Where that last term comes from. The replication splits the strip at F; the listed strip splits at K_0, which is a listed strike and so in general is not F. Expanding the logarithm about K_0 instead and taking risk-neutral expectations returns (22.5) with K_0 in place of F, plus the cost of having moved the split. With $x = F/K_0 - 1$, that cost is $2[\ln(1+x) - x] = -x^2 + O(x^3)$, the second-order remainder of a logarithm, and divided by T it is the subtracted term. Its size is bounded by $(\Delta K/K_0)^2/T$, a variance of about 0.00002 on the S&P's five-point grid near 4500, or about **0.005 VIX points** at a VIX of about 17.6, 24 days from expiration; Cboe carries

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it anyway because nothing guarantees a fine grid, and on a 25-point grid the bound is about 0.13 of a point. [TN22] derives it in full, states the exact correction the quadratic replaces, and measures it against the two larger errors that separate the published index from the variance swap rate.

Thirty days is a blend of two expirations. In general no option expires in exactly 30 days, so (22.7) is evaluated on the two nearest expirations that bracket it (the near term with more than 23 days to run, the next term with fewer than 37), and the results are interpolated *in total variance*, not in volatility:

$$\text{VIX} = 100\sqrt{\left[T_1\sigma_1^2\,\frac{N_2 - N_{30}}{N_2 - N_1} + T_2\sigma_2^2\,\frac{N_{30} - N_1}{N_2 - N_1}\right]\frac{N_{365}}{N_{30}}} \tag{22.8}$$

where σ_1^2 and σ_2^2 are (22.7) for the near and next expirations, N_1 and N_2 are the minutes from now to the near and next expirations, $N_{30} = 30 \times 1{,}440 = 43{,}200$ is the minutes in thirty days, $N_{365} = 525{,}600$ the minutes in a year, and $T_i = N_i/N_{365}$. The bracket is a linear interpolation of total variance to thirty days (variances over successive periods add; volatilities do not); the factor N_{365}/N_{30} annualizes it. The calculation counts minutes, not days, so the weights slide continuously through the session rather than stepping once a day. Weekly expirations have been eligible since 2014, which keeps the two brackets tight around 30 days; before that the window was wider and the interpolation did more work.

Example 22.4 Computing the index from a strip

Let the S&P 500 stand at 4500, with $r = 4.2\%$, options quoted on a five-point grid, and a downward-sloping skew: 17% at the forward, steepening to the downside. The near term expires in 24 days, the next in 31. The prices were generated with a 1.4% dividend yield, which the construction never sees.

Start with the forward. Equation (22.6) returns $F =$ **4508.29** for the near term, 8.3 points above the S&P's 4500 (the 24-day carry), so $K_0 = 4505$.

Then build the strip. The no-bid rule keeps 295 strikes, running from 3475 to 4945:

K	Option	ΔK	Price	$\Delta K e^{rT} Q/K^2$ ($\times 10^6$)
3500	put	5	0.060	0.024
4000	put	5	2.302	0.721
4300	put	5	20.540	5.570
4505	put/call avg.	5	78.347	19.355
4700	call	5	11.785	2.675
4900	call	5	0.207	0.043
Total over all 295 strikes				1020.478

Note how far the strip reaches on each side: 22.9% below the forward but only 9.7% above. The skew keeps deep puts bid long after the corresponding calls have lost their bids, so the strip is asymmetric before any truncation rule is applied.

Assembling (22.7):

$$\frac{2}{T}\sum \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) = 0.0310395, \qquad \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 = 0.0000081$$

so $\sigma_1^2 = 0.0310314$ and $\sigma_1 =$ **17.62%**. The correction moved the answer by 0.0023 VIX points, consistent with the bound above: the term is a safeguard rather than a working part.

The 31-day expiration gives $\sigma_2 =$ **17.82%**. The weights in (22.8) are ratios of the N, so their unit cancels; taken at a moment when the expirations are exactly 24 and 31 days away, $N_1 = 24 \times 1{,}440$ and $N_2 = 31 \times 1{,}440$ minutes, the weights are $(31 - 30)/(31 - 24) = 1/7$ and

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6/7, and

$$\text{VIX} = \mathbf{17.80}$$

closer to the 31-day figure than to the 24-day one, because 30 days is closer to 31.

Now thin the wings. Raise the price below which an option shows no bid, so the near-term strip, for example, runs from 3890 to 4835 instead of 3475 to 4945 (190 strikes rather than 295). VIX falls to **17.70**. No option changed its implied volatility; the index fell a tenth of a point because fewer options were quoted.

Three consequences follow from that experiment and from (22.7) and (22.8). The $1/K^2$ weighting makes VIX exceed the at-the-money implied volatility, because low strikes carry more weight and the skew makes them expensive. Apply (22.5) to the one-year skew of Figure 19.1 (standing in for a 30-day skew), interpolating between its plotted strikes and holding the smile flat beyond them, with zero rates, over strikes from 50 to 200: it returns 24.38% against an at-the-money implied volatility of 20.00%, while a flat 20% smile returns exactly 20.00%. The 4.38-point gap is the skew premium, and it widens as the skew steepens. On the strikes that skew actually quotes, 70 to 130, the answer falls to 21.99% and the flat control to 19.56%, so truncation costs something even with no skew at all.

The truncation rule makes VIX depend on quoting depth. The calculation stops including strikes when it meets two consecutive strikes with no bid, so on a quiet day the wings are thin, the sum is truncated early, and the index is measured on a narrower strip than on a busy one. That is worth a tenth of a point in Example 22.4, and more when volatility is high and the wings matter most. On August 5, 2024, before the US open, VIX printed 65.73 while S&P 500 futures were down about 3 to 4%: the formula was reading the midpoints of very wide pre-open SPX quotes. It closed at 38.57. That is Example 22.4's dependence on quotes rather than trades, at full scale and in the other direction: wide quotes inflated the midpoints the sum reads.

And the 30-day constant maturity is an interpolation. Two listed expirations bracketing 30 days are blended in total variance, so VIX refers to a synthetic maturity that no traded option has. As one expiration rolls out of the window and another rolls in, the weights in (22.8) shift, and around the roll the index can move for reasons that have nothing to do with volatility.

Together these make VIX a well-defined number that is not the price of anything that can be held. The strip that defines it today is not the strip that defines it tomorrow, so no position kept over time tracks it and it cannot be arbitraged against.

22.7 VIX FUTURES AND THE TERM STRUCTURE

A VIX futures contract is marked to market daily, like any futures contract, against a settlement price taken from trading at the close. Its final settlement, at expiration, is in cash against the **Special Opening Quotation** (SOQ): the VIX formula applied to the opening prices, set in a special opening auction on the settlement morning, of SPX options that expire exactly 30 days later, so no interpolation is needed; settlement falls on the Wednesday 30 days before that standard (third-Friday) SPX expiration. Like the index options of Section 10.3, the contract settles to a value that is computed rather than traded, and it can differ noticeably from both the previous close and the index a few minutes later. So the SOQ fixes the futures price on the final day, and the price converges to it. Before then the price is *not* tied to the index's current level, and the reason is worth being precise about.

A VIX futures contract is not a forward on VIX. Chapter 5's cost-of-carry argument requires that the underlying can be bought and stored, and VIX cannot be held. So there is no carry arbitrage linking the futures price to the spot index. A futures contract costs nothing to enter and is marked to market daily, so under the risk-neutral measure its price has zero expected growth (Section 13.10) and converges to the index at expiration: the futures price is the risk-neutral

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expectation of VIX_T, the index's value at the futures expiration T. Since VIX_T^2 is a 30-day variance rate and the square root is concave, Section 22.4's Jensen effect places the futures price below the square root of the forward variance swap rate, the fair variance rate for the 30 days that follow T (Section 22.12).

The practical consequence is the term structure. Because volatility mean-reverts (Section 19.6), the curve is usually upward sloping when spot volatility is low (which is most of the time) and inverts in stress: in Chapter 5's terms, contango in calm markets and backwardation in stressed ones. Mean reversion also makes the futures move less than the index: between the two curves of Figure 22.3 the index rises from 15.5 to 40 but the front contract only from 16.4 to 33, about two-thirds as far, and the five-month contract under a quarter as far. A hedge sized on spot VIX therefore under-hedges, and more so the further out the contract. Besides the issuers of Section 22.8's products, who hedge in the first two contracts, the curve's main traders are hedge funds positioning on its slope.

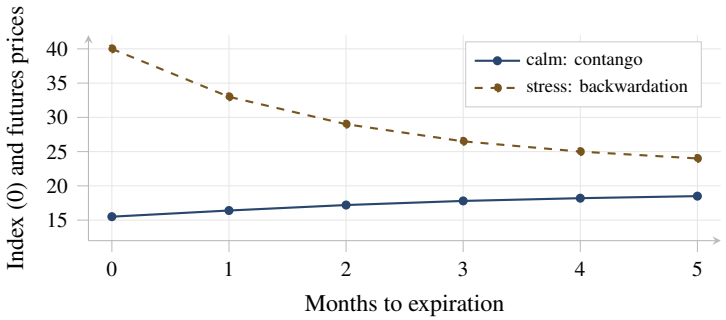


Figure 22.3: Representative VIX futures curves. Illustrative.

Example 22.5 What contango costs a long position

Using the calm curve of Figure 22.3: the front month is 16.4 and the second month 17.2.

A fund holding constant one-month exposure must roll: it buys the second month at 17.2, and one month later, if the curve has not moved, that contract has become the front month and is marked at 16.4.

$$\text{loss} = 17.2 - 16.4 = 0.8 \text{ points}, \qquad 0.8/17.2 = \textbf{4.65\% per month}$$

Compounded, $(1 - 0.0465)^{12} - 1 = -43.5\%$ **per year**, with the VIX index unchanged throughout.

This is Section 5.11's roll yield with no carry behind it, large because the VIX curve is steeper than most futures curves. It has two consequences, one for each side of the trade.

Long VIX futures products are structurally poor hedges to hold. An investor who buys volatility exposure as portfolio insurance and holds it for a year pays something like 40% of the position for the privilege, and the payoff must beat that before the hedge has done anything.

And short VIX futures products earn that roll as a carry, which is the volatility risk premium of Section 19.10 collected through a different instrument. Section 22.8 is about what happened to the people collecting it.

The premium and the roll are the same thing seen from two sides. Mean reversion alone would tilt the curve upward from a low spot level, but a futures price that merely forecast the index would cost a long holder nothing on average; the curve slopes up by *more* than that because sellers demand compensation for bearing the risk, and that excess is both the roll cost to the buyer and the compensation to the seller. Neither is a market inefficiency, and a strategy built on "harvesting contango" is not exploiting anything—it is being paid to take a risk that occasionally arrives.

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22.8 VIX EXCHANGE-TRADED PRODUCTS

Section 18.12 counted February 2018 among the episodes in which hedging moved the market. Here is the mechanism from the product's side.

An **inverse VIX product** is an exchange-traded note (a bank's debt) or fund whose value moves opposite, day by day, to an index of a rolling position in the first two VIX futures contracts: it rises 1% for every 1% that index falls, and falls 1% for every 1% it rises. It is in effect a short position in those contracts, packaged so that a retail investor can collect the roll carry of Example 22.5 without a futures account, and by early 2018 the two largest, XIV and SVXY, held about $3 billion between them. Offering $-1\times$ exposure *daily* means the product must rebalance every day to restore that exposure, exactly as the leveraged funds of Section 14.9 do.

Example 22.6 Why an inverse product must buy into a rally

A fund with net asset value (NAV) A maintaining $-1\times$ exposure is short futures with notional A. Suppose futures return R over the day.

The NAV becomes $A(1-R)$, so the new target short is $A(1-R)$.

The existing short has grown to $A(1+R)$ of exposure.

So the fund must buy

$$A(1+R) - A(1-R) = \mathbf{2AR}$$

of futures at the close. It buys as the price rises. The rebalance is destabilizing, and it is Section 18.12's short-gamma feedback in a different form.

With roughly $3 billion in inverse VIX products, the rebalancing works out as follows:

Futures move	NAV after	Buying required
+10%	$2.70bn	$0.60bn
+25%	$2.25bn	$1.50bn
+50%	$1.50bn	$3.00bn
+96%	$0.12bn	$5.76bn

The demand grows as the fund shrinks. At a 96% move the products have lost 96% of their capital and must buy $5.76 billion of futures (nearly twice their original size) into a market that is already moving against them.

The buying was due in the fifteen minutes before the futures close, when the market normally trades a small part of its daily volume. The rebalance was not a large order in a deep market; it was a large order in a thin one, announced in advance.

Figure 22.4 puts the two quantities on one axis, and the crossing is the whole problem. The purchase the fund owes the market rises with the move while the fund itself shrinks, and on any day the futures rise by more than a third the buy exceeds everything the fund has left. February 2018 lay beyond the point where those lines cross.

Case Study 22.2 February 5, 2018, from the volatility products

The S&P 500 fell about 4% that day—a bad day, not a historic one. VIX futures roughly doubled.

For a $-1\times$ product, a 100% move in the underlying futures is total loss: the NAV multiplier is $(1-R)$, and at $R = 1$ it is zero. The largest inverse product, XIV, had a term in its prospectus allowing early termination if the value the issuer published during the day (its intraday indicative value) fell more than 80% from the prior close. It did. The note was redeemed early, and holders received $5.99 a note, against more than $100 a few days earlier. SVXY, a fund holding the futures directly rather than a bank's note and

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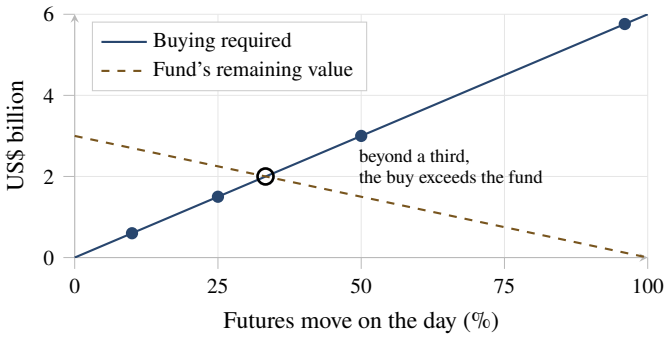


Figure 22.4: Daily rebalancing of a $-1\times$ inverse VIX product with \$3 billion of assets. Marks are the moves of Example 22.6.

carrying no such clause, lost about 90% on the same day but survived, and its sponsor cut the exposure to $-0.5\times$ three weeks later, where it remains. Three things about the episode deserve to be separated.

The product did what it said. A $-1\times$ daily inverse product loses everything on a 100% daily move, and the prospectus said so. This was not mis-selling in the narrow sense.

The rebalance amplified the move that destroyed it. Example 22.6 gives the size. The products had to buy into the close precisely because the price had risen, and their buying was a material fraction of the volume available. A product whose hedging requirement grows as it loses money is a product that helps arrange its own destruction.

The wider market was already fragile. Two years of low realized volatility had built large short-volatility positions well beyond the exchange-traded products: in the option-selling strategies of Section 19.10, in dealers' books, and in funds that size equity by its volatility (risk-parity and volatility-target funds), whose allocations fall mechanically when realized volatility rises. All of these were positioned the same way.

The general lesson is the one Section 18.12 drew from five separate episodes and Section 35.7 meets again in the autocallable market. Crowding plus mechanical hedging is the mechanism, and the instrument is incidental. 1987's portfolio insurance, 2018's inverse exchange-traded products, and 2024's snowball knock-ins are the same structure with different wrappers, and the recurring feature is that the hedging demand is a known function of the price, in 2018 a published one, calculable in advance by anyone who cared to.

22.9 THE PREMIUM, SEEN THROUGH EACH INSTRUMENT

Section 19.10 reported that implied volatility exceeds subsequently realized volatility by two to four points on equity indices. In variance-swap form that premium is measured as VIX^2 against the next 30 days' realized variance, as Carr and Wu (2009) measured it, and it varies over time. The three instruments of Section 22.1 all collect that premium, and they do not collect the same amount. The difference is a useful check on whether one understands what each contract is, and Example 22.7 applies it.

Example 22.7 One premium, three numbers

Suppose one-year implied volatility is 18% and realized comes in at 15%.

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A short delta-hedged straddle earns, by Section 22.1, the difference between implied and realized *variance*, gamma-weighted along the path:

$$18^2 - 15^2 = 324 - 225 = 99 \text{ variance points (squared volatility points)}$$

but the realization depends on where the underlying spent its time. If the stock spends the year far from the strike, the position has little gamma when the moves occur and collects a fraction of the 99.

A short variance swap at 18% earns exactly 99 variance points. Per unit of vega notional, by (22.2) the short position's payoff is $(K_{\text{var}}^2 - \sigma_R^2)/(2K_{\text{var}})$, here

$$\frac{18^2 - 15^2}{2 \times 18} = \textbf{2.75 volatility points}$$

with no dependence on where the moves occurred. It is less than the 3-point gap in volatility because the payoff is convex in σ_R (Section 22.4): the short gains less when realized volatility falls short of the strike than it would lose if it overshot by the same amount. The swap collects the straddle's quantity cleanly, which is the reason the contract exists.

A short VIX futures position earns something else. It collects the roll of Example 22.5 (0.8 points a month on the illustrative curve), which is a claim about the *shape of the curve*, not about the gap between one-year implied and one-year realized volatility. The two are related, but they are different contracts on different quantities and can move apart.

Instrument	What is collected	What can go wrong
Delta-hedged straddle	Implied − realized variance, gamma-weighted	Path: the gamma may not be where the moves are
Variance swap	Implied − realized variance, exactly	Convexity: the loss is quadratic
Short VIX futures	The roll down the curve	The curve inverts, and does so before realized volatility rises

The third row contains the trap. The VIX curve inverts on the *expectation* of volatility, which is to say immediately, while realized volatility rises only as the moves actually occur. A short VIX futures position therefore takes its loss at once, while a short variance swap's settlement loss accrues only as the moves occur; its mark-to-market also loses as implied variance rises, but the payoff waits for the realized path. Anyone treating the two as interchangeable will find the correlation between them is high in calm markets and much lower in exactly the week it matters.

22.10 WHAT GOES WRONG IN A VARIANCE SWAP BOOK

The replication of Section 22.3 is exact under assumptions, and a dealer's business is the gap between those assumptions and the world. Four gaps matter.

Jumps. The log-contract argument assumes a continuous path. A gap of log size J adds J^2 to realized variance (before annualization), while the replicating portfolio collects $2(e^J - 1 - J)$, which differs by about $J^3/3$ and falls short on a downward gap. The replication under-hedges precisely on the days that matter, and this is the residual that Section 21.5 said cannot be hedged by any position taken beforehand.

Dividends. Realized variance is computed from price returns, and a price drops on the ex-dividend date. Unless the contract adjusts for it (index contracts usually do not; single-stock contracts often add back the dividend actually paid), a large special dividend registers as volatility that never happened.

Single stocks. Section 22.5 recorded that single-stock variance swaps, capped from the outset, nearly stopped trading after 2008. The reason is the convexity of Example 22.3 applied to an underlying that can gap 40% on an earnings miss or a takeover bid: the seller's loss, even capped, is quadratic up to the cap in a move whose distribution has a much fatter tail than an

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index's. Dispersion desks that would naturally use single-stock variance often use straddles or other options on the constituents instead, because the capped swaps trade thinly and the cap cuts off the payoff on the large single-stock moves the component leg is long. That substitution reintroduces exactly the path dependence the instrument was designed to remove.

The strip is not the contract. The dealer holds a fixed portfolio of options; the client holds a claim on realized variance. If the underlying moves far enough that the strip's strikes are all deep in or out of the money, the replication's gamma no longer matches the log contract's, and the hedge degrades exactly when realized variance, and so the payoff owed, is largest. This is the same truncation issue as Example 22.1, now as a running risk rather than a pricing error.

None of these is a flaw in the theory. Each is a place where a contract defined to be model-free requires a judgment, and a variance-swap desk is in the business of pricing those judgments.

22.11 DISPERSION

Section 16.2 named dispersion traders and Section 19.7 gave the arithmetic: index and single-stock implied volatilities together imply a correlation. The dispersion trade of Example 19.4, selling index volatility against single-stock volatility, is a position in that correlation, and with variance swaps it can be constructed cleanly.

Take the numbers of Example 19.4, its implied volatilities taken as one-year variance swap strikes: 50 equally weighted stocks at 30%, the index at 18%, implying a correlation of 34.7%; the 2.6-point gap Example 19.4 found at a realized 25% is 2.38 points as a variance payoff (first row below). The weight on the component leg follows from Example 19.4's equal-weight form of (19.3). A parallel move in component variance σ_c^2 moves index variance σ_I^2, at fixed correlation, by the factor $\sigma_I^2/\sigma_c^2 = (0.18/0.30)^2 = 0.36$, so 0.36 units of variance notional spread equally over the 50 component swaps against 1 unit of index variance makes the position neutral to volatility and exposed only to correlation. In general the weight is $(K_I/K_c)^2$, the square of the index strike K_I over the component strike K_c.

Figure 22.5 plots the resulting profit against realized correlation, and the line is the point of the structure: it crosses zero at the implied 34.7% whatever the component volatility, and is linear in correlation with a slope set by component variance. Example 22.8 reads four outcomes off it and adds a calm one.

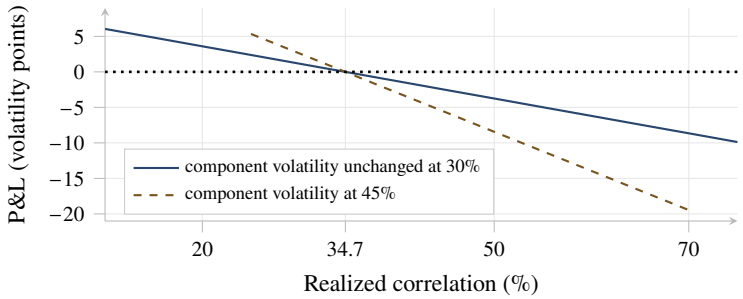


Figure 22.5: Profit on a variance-neutral dispersion trade against realized correlation.

Example 22.8 What dispersion earns and what it loses

The position is short 1 unit of index variance at 18% and long 0.36 units of component variance at 30%.

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Component vol	Realized corr	P&L	
30%	25.0%	+2.38	correlation below implied
30%	34.7%	0.00	correlation realizes at implied
30%	45.0%	−2.53	correlation rises, volatility unchanged
20%	20.0%	+1.60	calm: everything falls
45%	70.0%	**−19.46**	crash: volatility and correlation both jump

(P&L in volatility points per unit of index vega notional.)

The second row is the check. When realized correlation comes in exactly at the implied level, the trade makes precisely zero, whatever volatility does, as long as the components realize a common volatility. No parameter was tuned to produce it; it follows from the weighting, and it confirms that the position is a pure correlation bet.

The last row is the trade. The first row, correlation ten points below implied, earns 2.38 points; the crash loses 19.46. The ratio is one to eight, and the loss arrives in the same weeks as everything else in the portfolio.

Two causes combine. The correlation loss is scaled by realized component variance, and in a crash both rise together: at an unchanged 30% volatility, 70% correlation costs 8.65 points; at the 45% that accompanies it, 19.46. And correlation starting at 34.7% has nearly twice as far to rise as to fall, so the largest possible moves are the ones that run against the short index leg.

This is why the trade is described as "earning steadily and losing in blocks," and why it is run by desks that can size it against a capital base rather than by funds that must report monthly. Cboe publishes implied correlation as an index (COR1M, COR3M). In July 2024, with index returns concentrated in a few very large stocks, the three-month index fell to a record low and the trade had become crowded; when volatility spiked on August 5 (Section 22.6), correlation rose with it, in the direction of the last row.

Correlation swaps. A correlation swap pays realized average pairwise correlation against a strike directly, which sounds cleaner and is used less. The reason is replication: dispersion is built from variance swaps that can themselves be replicated from options, whereas a correlation swap cannot be replicated, and the dealer who writes one hedges it only approximately, with a dispersion position whose payoff is a variance-weighted proxy for the same correlation. The market trades the replicable proxy rather than the clean instrument, which is the same preference that made variance rather than volatility the standard contract. The buyers are mostly structured-product dealers, short correlation from worst-of notes (Section 34.11); the writers are hedge funds and dispersion desks already holding that view through variance.

22.12 FORWARD VARIANCE AND OPTIONS ON VOLATILITY

Section 21.8 observed that rough-volatility models specify the forward variance curve, today's fair variance rate for each future window, directly. That curve is a traded object, and this is where it trades.

A **forward-starting variance swap** contracts on variance over a future window, and it is replicated by a spread of two spot-starting variance swaps: long the longer, short the shorter, weighted by their maturities. Since both legs are replicable from options, so is the forward: the whole forward variance curve is spanned by today's vanilla surface, and it needs no storable underlying, which Chapter 5's forward prices do. Dealers use the forward swap to hedge the forward-volatility exposure of the cliquets and autocallables they issue (Section 34.12, Chapter 35), and volatility funds to take term-structure positions. Example 22.9 works one through.

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Example 22.9 Reading the forward variance off two spot swaps

Six-month variance swaps are quoted at 18% and one-year at 20%. Total variance to each date is TK^2_{var}: $0.5 \times 0.0324 = 0.0162$ and $1 \times 0.0400 = 0.0400$. The variance over the second half-year is the difference, spread over the half-year it covers:

$$\sigma^2_{\text{fwd}} = \frac{1 \times 0.0400 - 0.5 \times 0.0324}{1 - 0.5} = \frac{0.0238}{0.5} = 0.0476, \qquad \sigma_{\text{fwd}} = \mathbf{21.82\%}$$

The forward sits above both spot strikes, as a forward rate sits above both spot rates on an upward-sloping curve in Chapter 4, and for the same reason: a two-point rise in the term structure over a horizon that is only half the maturity has to be made up entirely in the later half. The check is the telescoping identity: the two half-year variances average to the one-year, $(0.0324 + 0.0476)/2 = 0.0400$.

Long one unit of the one-year swap and short half a unit of the six-month swap, in variance notional, is long 0.5 units of forward variance; scaled to one unit, long 2 against short 1, which is what "weighted by their maturities" meant.

What is *not* spanned is anything nonlinear in future variance. VIX futures (a square root of variance), options on realized variance and on VIX, and the caps of Section 22.5 all require a model of how variance moves, not merely its expectation. This is the boundary Chapter 21 drew.

VIX options are the largest such instrument. They are European and settle in cash to the same SOQ as the futures, and their forward is the futures price for the same expiration, not spot VIX, so put–call parity runs against the futures. Pricing them requires a joint model of the S&P 500 and of the volatility surface from which VIX is computed. Their implied volatilities carry an upward skew, the reverse of the equity skew, because VIX spikes up and not down, and the market pays more for the right tail of a quantity that spikes. Tail hedgers buy the out-of-the-money calls as crash protection; volatility sellers and dealers write them. The VIX formula applied to a strip of VIX options gives VVIX, the implied volatility of VIX itself. Fitting the SPX smile and the VIX smile with one model is the joint calibration problem of Section 21.8, and the steep VIX skew is what makes it hard: it demands a volatility of volatility the SPX smile alone does not reveal.

22.13 WHO TRADES WHAT

Buyers of volatility are the protection buyers of Section 16.2 using a different instrument. A pension fund buying VIX calls or entering a long variance swap is buying convexity against its equity book. It pays the premium of Section 19.10 and the roll of Example 22.5, and it does so knowingly: the alternative is holding less equity.

Sellers of volatility are collecting that premium. They divide into those who can hold the position through a drawdown (dealers with capital, insurers, and endowments) and those who cannot, which is most of the rest. Allianz Global Investors' Structured Alpha funds show a seller less hedged than it said: they sold index options against crash puts their investors were told they held, but the managers had bought cheaper puts further out of the money, and in March 2020 funds that had run more than $11 billion lost over $7 billion. The firm's US arm pleaded guilty to securities fraud in May 2022 and paid about $6 billion.

Dealers intermediate and end up long or short depending on what clients trade, hedging with the vanilla strip of Section 22.3; their business is the *difference* between that strip and the contract (Section 22.10).

Dispersion desks trade correlation, as Section 22.11 described, and are the natural counterparty to index protection buyers: the index put demand that steepens the index skew is what makes implied correlation high enough to be worth selling. Worst-of note issuance is a

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second source of that flow: the investors are long correlation, so the issuing dealers are short it (Section 34.11) and buy it back by buying index volatility against single-stock volatility, which lifts implied correlation further.

Volatility-target and risk-parity funds do not trade volatility products at all, but they are among the largest volatility *exposures* in the market, because their equity allocation is a decreasing function of realized volatility. A fund targeting 10% volatility holds equity in proportion $10\%/\sigma_R$, so realized volatility doubling from 12% to 24% halves its equity weight from 83% to 42%; a hypothetical $100 billion of such funds then sells about $42 billion of equity. They are mechanically short volatility without holding a single option, and Section 22.8's snapshot showed what that means when realized volatility jumps.

Investors, taken together with their equity, are structurally short volatility, and the premium is the price of shedding that risk. What the episodes in this chapter show is that the capacity to bear it is smaller than the appetite to collect it, and the gap is discovered at inconvenient moments.

22.14 WHERE THIS CAME FROM

The instruments in this chapter rest on a result that needs no model at all. Breeden and Litzenberger (1978) showed that the option chain reveals the terminal distribution outright, and Carr and Madan later showed that any payoff written on that distribution can be replicated from options. Neuberger's log contract (1994), the payoff whose delta-hedged profit is exactly realized variance, is what variance swaps were built on, and Demeterfi, Derman, Kamal and Zou (1999) wrote the replication out in the form desks use, the source of Section 22.3's $1/K^2$ weighting. The CBOE rebuilt VIX on the same replication in 2003, which is why the index is a variance calculation rather than an average of implied volatilities, and Carr and Wu (2009) then measured the premium the trade earns, Section 19.10's premium in variance terms.

SUMMARY

Volatility can be owned three ways (delta-hedged options, variance swaps, VIX futures), and they are not substitutes. The first pays realized minus implied variance weighted by gamma, so it depends on where the underlying moved; the second pays realized variance exactly; the third is a claim on the level of a computed index.

A variance swap pays $N_{\text{vega}}(\sigma_R^2 - K_{\text{var}}^2)/(2K_{\text{var}})$ on a realized variance the contract defines. It can be replicated statically from out-of-the-money options weighted $1/K^2$, because delta-hedging a log contract earns realized variance on every path and the log contract's payoff is replicated by that strip. Variance rather than volatility is traded because variance is replicable; the variance swap strike exceeds the volatility swap strike by an amount that grows quickly with the volatility of volatility, so it is not a volatility forecast. The same convexity makes the seller's loss quadratic: after 2008 the single-stock market, capped from the start, shrank, and capped index contracts appeared too; a capped contract needs a model, and March 2020 showed what a cap is worth.

VIX is the 30-day variance swap rate on the S&P 500 as Cboe's rule computes it: above at-the-money implied volatility because the $1/K^2$ weights load on the low strikes the skew makes expensive, dependent on quoting depth because of the truncation rule, and an interpolation between two expirations. It is a well-defined number that is not the price of anything that can be held. VIX futures are not forwards, because VIX cannot be held; rolling a long position on a contango curve loses steadily, at a rate that compounds over a year to a large share of the position, and that cost to the buyer is the premium to the seller.

An inverse VIX product with net assets A must buy $2AR$ of futures when futures return R, so its hedging demand grows as it loses money; on February 5, 2018 that demand overwhelmed

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the liquidity at the close. Dispersion sells index variance against component variance weighted $(K_I/K_c)^2$, makes zero when correlation realizes at the implied level, and loses many times its normal gain in a crash, because correlation rises when everything falls and the loss on it is scaled by component variance, which rises with it. The forward variance curve is spanned by today's vanilla surface; anything nonlinear in variance is not, and needs the models of Chapter 21.

FURTHER READING

Demeterfi, Derman, Kamal and Zou's paper on variance swaps is the original practitioner derivation of (22.5); it works the replication through in a way this chapter compresses. Carr and Lee survey what can and cannot be replicated in volatility, and their survey is the place to go for the volatility-swap adjustment stated properly rather than approximated. Cboe's VIX methodology documents the construction including the truncation rule and the roll, and is worth reading precisely because the index's peculiarities are all in the mechanics rather than the theory. On February 2018, the Bank for International Settlements' analysis in its March 2018 Quarterly Review is the standard account; the mechanism of Example 22.6 is arithmetic, and the question it addresses is how much of the day's move the rebalance actually caused. Bergomi's book (2016) is the practitioner reference for forward-variance models and for options on variance and on VIX.

PRACTICE QUESTIONS

22.1 A variance swap is struck at 20% with \$100,000 of vega notional. Compute the variance notional and the payoff at 35% realized, compare it with what a linear-in-volatility contract would give, and explain what determines the gap between a variance and a volatility swap strike.

22.2 Explain why delta-hedging a log contract earns realized variance regardless of the path, what feature of the log contract's gamma is responsible, and why the replication strip weights options by $1/K^2$ rather than equally.

22.3 Using Example 22.1, state whether widening the strike range or narrowing the strike spacing is the more effective way to improve a replication at low volatility, and explain why the answer changes at high volatility.

22.4 Six-month variance swaps are quoted at 25% and one-year variance swaps at 22%. Compute the forward volatility for the second half-year, explain why it lies below both spot strikes, and state the positions in the two swaps, in variance notional, that make up one unit of forward variance.

22.5 Explain why variance swaps, single-stock ones above all, are now often written with caps, and what the cap does to the replication argument.

22.6 Explain why VIX exceeds the at-the-money implied volatility of 30-day S&P options, and why a VIX futures contract is not a forward contract on VIX in the sense of Chapter 5.

22.7 Using the curve of Figure 22.3, compute the monthly and annualized roll cost of a constant one-month long VIX futures position, and state who receives it.

22.8 A $-1\times$ inverse VIX product has \$2bn of NAV. Compute the rebalancing trade required if futures rise 30% in a day, state its direction, and explain why the requirement grows as the product loses money.

22.9 Explain why a dispersion trade weighted $(K_I/K_c)^2$ makes exactly zero when realized correlation equals implied correlation, whatever the components' common volatility, and

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why it loses so much more in a crash than it earns in a normal quarter.

22.10 Implied volatility is 22% and realized comes in at 18%. Compute what a short variance swap earns in volatility points, and explain why a short delta-hedged straddle need not earn the same and why it typically earns less.

22.11 A special dividend of 8% is paid on a stock during the life of a variance swap that does not adjust for dividends. Explain what happens to measured realized variance and who gains.

22.12 Explain why dispersion desks often use straddles rather than variance swaps on single stocks, and what they give up by doing so.

FURTHER QUESTIONS

22.13 Example 22.1 finds that widening the strike range beyond 50–200 buys nothing at 20% volatility, while changing the spacing moves the result. Derive the truncation error and the discretization error, show how each scales with volatility and maturity, and determine the volatility at which they are equal for a one-year contract on strikes from 50 to 200 spaced 5 apart, with the forward on a listed strike.

22.14 Section 22.5 argues that capping a variance swap destroys its static replication and therefore requires a model. Decompose a capped variance swap into a variance swap and a call on variance, explain what information about the process the call requires that the swap does not, and connect this to Section 21.11's measurement of model risk.

22.15 Example 22.5 finds a 43.5% annual roll cost and asserts that this is compensation rather than an inefficiency. Construct the argument that a systematic short VIX futures position is a risk premium, then construct the argument that it is a crowded trade earning a rent, and state what evidence would distinguish them.

22.16 Example 22.6 shows that an inverse product's rebalancing demand is $2AR$. Derive the corresponding expression for a $+2\times$ leveraged long product, explain why both signs of leverage are destabilizing, and discuss whether a product rebalancing at a random time of day rather than at the close would be safer or merely less predictable.

22.17 Section 22.13 claims that volatility-target funds are mechanically short volatility without holding options. Formalize this: write the fund's equity allocation as a function of realized volatility, derive its sensitivity to a change in realized volatility, and express that sensitivity in units comparable to an option's vega. Then discuss what this implies about measuring the market's aggregate short-volatility position.

22.18 Example 22.8 shows a one-to-eight ratio between the first row's gain and the crash loss. Derive the general condition on the implied correlation under which a variance-neutral dispersion trade has positive expected value, given a joint distribution for realized correlation and component volatility, and explain why a desk might rationally run the trade even when that condition fails.

22.19 Example 22.9 reads a forward variance off two spot variance swaps. Repeat it for a curve that is *inverted* (six-month at 30%, one-year at 24%), and show that the forward strike falls below both spot strikes. Then find the six-month strike at which the implied forward variance would be zero, and explain what a quote above that level would mean and how it would be arbitraged.

22.20 Section 22.12 states that the whole forward variance curve is spanned by today's vanilla surface with no storable underlying required. Explain why the variance case needs no storage, identify the assumption that makes it work, and state what fails if that assumption is relaxed.

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